# Heat is not temperature

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Distinguish temperature from energy transferred because of a temperature difference.

**SUCCESS CRITERIA - I can...**

1. Compare temperature with thermal energy.
2. Identify the direction of spontaneous heat transfer.
3. Explain why amount and material matter.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

For fixed GM, launch radius becomes 9 times as large. Compare ideal escape speeds and name the equation.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $Q = mc\,\Delta T$.
For heating without a phase change: Q is transferred heat, m is mass, c is specific heat. Heat and temperature are different quantities.

**Worked reasoning:** A cup and a bathtub can both be at 40°C. Equal temperature does not imply equal internal energy: amount and type of material also matter.

**Your turn:** Draw equal temperature indicators but different amounts of water. Predict heat-flow direction if the water samples are brought into thermal contact at the same temperature.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** ① ② ③

## 03 Read and interpret the model

A cup and a bathtub contain the same substance at 40°C. Are their temperatures equal? Must their total internal energies be equal?



Cup: 40°C Tub: 40°C

Each outline represents the labeled sample or container.

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

Two containers hold the same liquid at the same temperature. Container B has three times as much liquid by mass. Compare the energy needed to warm each by the same temperature change. Label the model and state your assumptions.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** ① ② ③

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Continue down each column.

## 05 Explain your reasoning

A student says “cold flows into a hot drink.” Rewrite the explanation in terms of energy transfer.

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A point-mass pendulum drops 0.8 m vertically from rest. Use g=10 and neglect drag. Find bottom speed. Show the energy changes and transfers; label units.

**Initial** **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** 1 2 3

## 07 Exit check

Can equal-temperature objects still contain different internal energies? Give two reasons.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

Two equal masses of water start at 20°C and 60°C and mix in isolation without phase change. Find final temperature, then explain why this averaging rule would fail for unequal masses.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** 1 2 3